

Digital Safety Transparency Report - Microsoft Learn

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Introduction

Microsoft Learn is a customer-facing website that helps our customers gain skills with Microsoft technologies through a variety of learning solutions, some of which include user-generated components. There are four primary areas where these components exist:

- **Q&A** on Microsoft Learn is an integral part of the Microsoft Support ecosystem, designed to assist users in resolving technical issues and enhancing their experience with Microsoft products. A vital aspect of Q&A is the facilitation of user-to-user communication. This interaction allows users to help each other by sharing their experiences and solutions to common problems. Users can ask questions about Microsoft solutions and receive answers from other users, or they can review questions that have been posted and provide answers based on their own experiences and expertise. The scalability of this model is significant, as many users can benefit from existing content and discussions, reducing the need for repetitive queries and enhancing the overall efficiency of the support system.
- **Collections and Plans** are user-curated lists of existing Microsoft Learn materials. Users can provide a title, description, and goals which would then be visible to other users who visit them.
- **Profiles** capture a user-generated username and user display name that may be publicly visible.

Data in this document reflects an aggregate of reporting across these experiences.

Contents

This DSA Transparency Report is responsive to the obligations under DSA Article 15(1), Article 24(1), and Article 42(2)-(3). This Report provides information regarding the following topics as they pertain to the European Union:

- Content Moderation at Microsoft's Initiative
- Content Moderation & Automated Systems
- Content Moderation following a User Report
- Content Moderation Appeals

- Government Requests

Content Moderation at Microsoft's Initiative

Summary of the content moderation engaged in at the providers' own initiative

Microsoft provides the information below in response to DSA Article 15(1)(c). This section reports data regarding content moderation Microsoft Learn engaged in on its own initiative, absent a user report.

This section describes activities Microsoft Learn undertakes to address harmful content or information in violation of Microsoft Learn's terms and conditions, which can be found [here](#).

There are four areas of Microsoft Learn where content moderation takes place: Q&A, Collections, Plans, and Profile.

When Microsoft Learn flags content through automated detection, the service may restrict the content owner's ability to edit or share that content while the content awaits human review.

Q&A uses both automatic detection and human moderation to keep the platform safe. Moderators have internal built-in tools to manage the content and user activity.

The following harm management roles are in place:

Role	Qualification	Permissions
Admin	Microsoft FTEs	Thread/Profile level harm management Suspend/Unsuspend moderators and users
Microsoft Moderator	Microsoft FTEs	Thread/reply level harm management Suspend users
Volunteer Moderator	Trusted and tenured external community members	Thread/reply level harm management Suspend users

Q&A also has automated detection for spam content. Spam in the context of Q&A refers to unsolicited behavior. Examples on spam range from posts sharing credit card details or phone numbers for lucrative offers or posts soliciting help for personal matters. In instances of confirmed violations, Microsoft will implement appropriate measures including content restriction, modification or removal, and potential community account suspension, preventing the offending party from generating new content. In cases of severe violations, particularly posted content related to or participating in the distribution of child

sexual abuse material, Microsoft will terminate the content owner's Microsoft account in alignment with established Microsoft policies.

Collections and Plans also incorporate automated spam detection. When violations are confirmed, Microsoft will take appropriate actions such as restricting, modifying, or removing content, or deleting the violator's profile to prevent further content generation. In cases of severe violations, especially the distribution of child sexual abuse material, Microsoft retains the right to terminate the account of the content owner in accordance with established Microsoft policies.

Profile uses automatic detection to keep the platform safe. When violations are confirmed, Microsoft will take appropriate actions such as restricting, modifying, or removing content, or deleting the violator's profile to prevent further content generation. In cases of severe violations, especially the distribution of child sexual abuse material, Microsoft retains the right to terminate the account of the content owner in accordance with established Microsoft policies.

In all four of these areas, violating content is blocked from sharing or is removed, and, in the case of egregious violations such as sharing child sexual abuse material, the Microsoft account of the content owner may also be terminated in accordance with Microsoft's published policies.

Meaningful and comprehensible information regarding content moderation engaged in at the provider's own initiative

Microsoft Learn uses automated content detection on text and images when it is shared by a user account.

Microsoft Learn leverages automated content detection to detect known child sexual abuse material (CSAM) and terrorist content using both PhotoDNA and MD5 hashes.

PhotoDNA is a perceptual hashing technology, which is used to identify visually identical and near identical images and video. To do so, PhotoDNA creates a unique digital signature (known as a "hash") of an image or selected frames in a video and compares it against signatures ("hashes") of previously identified CSAM and terrorist content images or video. PhotoDNA is not facial recognition software and cannot be used to identify a person or object in an image.

For CSAM, PhotoDNA is used to detect content that has previously been determined to sexualize a child based on a classification scale widely adopted across industry. The service draws from an internal database of Microsoft-created hashes for CSAM content, as well as hashes provided by trusted third-party organizations.

For terrorist content, PhotoDNA is used to detect content that has previously been determined to be violent extremist and terrorism content consistent with Microsoft policies and which has been hashed for inclusion in an internal database of Microsoft-created hashes, as well as hashes provided by the Global Internet Forum to Counter Terrorism (“GIFCT”).

Message-Digest Algorithm (“MD5”) is a broadly used cryptographic hashing technology, which is provided as part of standard programming libraries and is used to determine if two items of content are identical. Microsoft currently deploys MD5 in conjunction with PhotoDNA to identify CSAM and terrorist content.

Product Data: Microsoft Learn

Policy Violation	Number of measures taken at the provider's own initiative	Number of measures taken after detection with solely automated means	Visibility restriction Removal
Pornography or Sexualized Content	15	15	15
Scope of Platform Services	14	14	14

Content Moderation & Automated Systems

Qualitative description of the automated means

Microsoft provides the information below in response to DSA Articles 15(1)(e) and 42(2)(c).

There are several automatic tools that scan for harms on Microsoft Learn:

1. **Auto suspend users:** This service auto-suspends users after three strikes of posting spam
2. **Keyword spam list:** This service blocks keywords that are added to the spam list
3. **Captcha 3rd party service:** This service helps detect bot profiles
4. **Allow list for images from specific domains:** This service allows for images from specific domains, reducing our image related spam

The specifics of the technical implementation of PhotoDNA and MD5 hash matching are covered above.

The PhotoDNA distance threshold, which determines how “close” the Euclidian distance between two hashes needs to be in order to consider them a match, was selected based on the largest distance for which there were no false positives as determined during

PhotoDNA's initial validation, performed in partnership with the National Center for Missing and Exploited Children.

Given that MD5 is solely capable of identifying exact matches, there are no parameters set for its use.

Microsoft Learn performs automated detection by generating PhotoDNA and MD5 hashes for the content to be scanned and comparing the resulting hash against the hashes of known bad content (as earlier described). If a match is found, the suspected CSAM or terrorist content is verified through human review, with the following exceptions:

- If additional content that is awaiting review has the same hash, only one such instance will be reviewed by a human reviewer and the result of that review will be applied to all content pending review with that same hash.
- If there have been at least ten earlier human reviews that confirmed CSAM content from a given content owner within the previous day, then all additional content from that same content owner that matches a CSAM hash will be automatically actioned on as confirmed CSAM without additional human review.

If content is confirmed to be violative as described above, Microsoft Learn will block sharing or remove the content and may terminate the account in accordance with Microsoft's published policies. The service also notifies the content owner of the enforcement action and, if the content appears to be CSAM, reports it to the U.S. National Center for Missing and Exploited Children.

Qualitative description of indicators of accuracy and possible rate of error of automated means

There are two primary contributors to the effectiveness of hash matching systems that leverage PhotoDNA or MD5: the correctness of the hashing algorithm itself (the technology) and the management of the set of hashes against which content is matched (the content). The technology itself is content agnostic.

The correctness of the technology is ensured by validating that any updates do not alter its results when tested against a known set of inputs (i.e., we confirm that the hash calculations return the same result).

Hashing algorithms leverage databases of reference hashes of previously identified CSAM and terrorist content images or videos when assessing potentially violative content to order to find copies of the same content. Given that PhotoDNA and MD5 hash values are guaranteed to be stable, the precision and recall of these algorithms is, by definition, 100% for a set of previously identified violative hashes.

The following information relates to how the performance of the content (i.e. reference hashes) is tested and monitored. This is separate and distinct from the performance of the technology and is impacted by match distances used and the quality and consistency of human reviews both by Microsoft and by the third parties that generate hashes that Microsoft leverages.

Microsoft's reference hashes are updated and maintained in the following ways:

- Human review improves the accuracy of the underlying hashes by considering what the reviewer determined.
- In addition, in the case of PhotoDNA, hashes that are very similar are grouped to ensure they provide the same result.

As an indicator of accuracy, we have reversed 158 instances of Terrorist and CSAM enforcement actions. There is no direct measure of recall, though it should be noted that the previously identified hashes represent a small but unknown fraction of all violative content.

As noted previously, human review is not leveraged in cases where it would have no impact on the outcome of the review process (such as the prevention of multiple reviews of duplicate content) or would have no significant impact on the resulting enforcement actions (such as when the content owner's account has already been terminated for repeated confirmed violations). The measures of precision and recall of such actions would be equivalent to that of the underlying hash sets as described above.

Specifications of the precise purposes to apply automated means

Microsoft performs automated detection for child sexual abuse material (CSAM) because Microsoft is committed to protecting children from online harm. We do not allow the exploitation of, harm, or threat of harm to children on our services. This includes banning the use of our services to further child sexual exploitation and abuse (CSEA). When we become aware of content containing apparent child sexual exploitation and abuse, Microsoft reports the content to the National Center for Missing and Exploited Children (NCMEC).

CSEA is any content or activity that harms or threatens to harm a child through exploitation, trafficking, extortion, endangerment, or sexualization. This includes the sharing of visual media that contains sexual content that involves or sexualizes a child. CSEA also includes grooming, which is the inappropriate interaction with children by contacting, private messaging, or talking with a child to ask for or offer sex or sexual

content, sharing content that is sexually suggestive, and planning to meet with a child for sexual encounters. A child is anyone under 18 years old.

Microsoft performs automated detection for terrorist content because we want to protect our users from illegal and harmful content and conduct. We also recognize that we have an important role to play in preventing terrorists and violent extremists from abusing online platforms. We do not allow content that praises or supports terrorists or violent extremists, helps them to recruit, or encourages or enables their activities. We look to the United Nations Security Council's Consolidated List to identify terrorists or terrorist groups. Violent extremists include people who embrace an ideology of violence or violent hatred towards another group.

In addressing terrorist and violent extremist content, we also work to ensure that people can use our services to talk about terrorism or violent extremism, share news or research about it, or express opposition to it.

Microsoft chooses not to perform human review on content with a hash that matches the hash of another content item that was already awaiting human review, because the matching hash value confirms that additional reviews of that same content provide no additional benefit.

Microsoft chooses not to perform human review of additional suspected CSAM content from a given content owner after at least ten content items from that owner have been confirmed by human review to be CSAM within the previous day because, based on published policy, the associated account would have been terminated after the first such confirmation and, thus, the marginal value of reviewing additional violations is limited with respect to informing enforcement actions.

In both cases, the elimination of these low-value reviews reduces the frequency with which human reviewers are exposed to harmful content that could negatively impact their wellness.

Safeguards applied to the use of automated means

Microsoft leverages a broader human review process in conjunction with hash-based automated content detection. A matched image is directed to a human moderation queue (depending on the harm type of the matching reference hash), where one or more human reviewers may independently review the content to confirm that the content is violative before enforcement action is taken.

In cases where no human review occurs because content with the same hash was already in the process of being reviewed or the owning account had already been terminated as a

result of at least ten other content items that had been reviewed by humans and confirmed to be CSAM, the impacted content owner can submit an appeal for the return of the content.

In the case of a user whose account is terminated following the detection of CSAM by automated means, the impacted user can submit an appeal to have the account returned. As an additional safeguard, Microsoft monitors the frequency with which such appeals result in the account suspension being reversed.

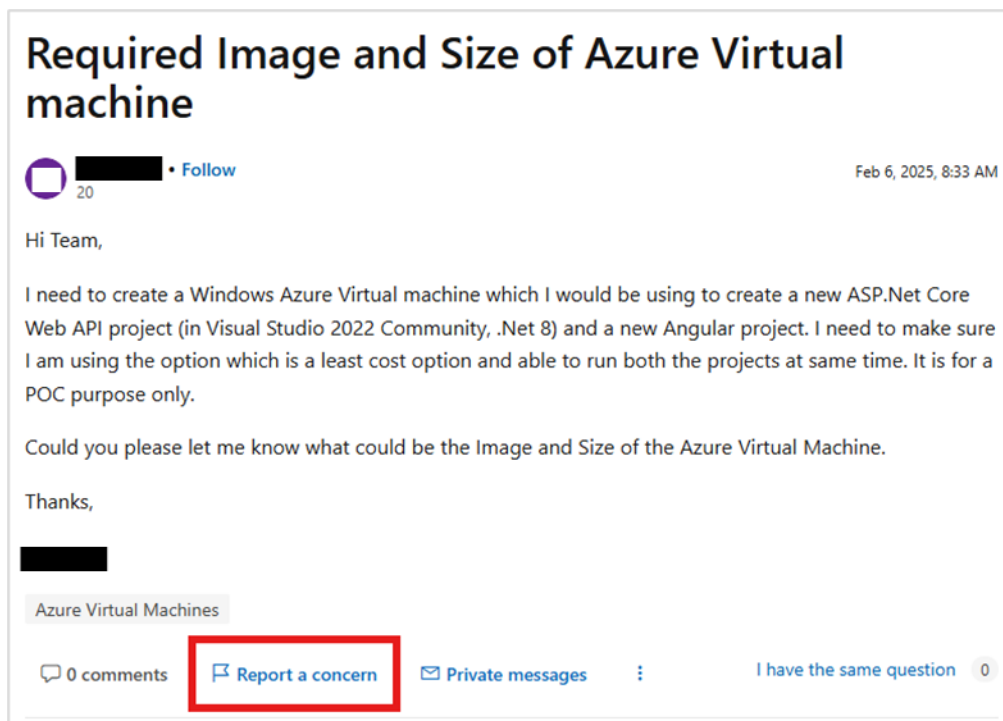
	Microsoft Learn
Number of measures not taken by automated means	29

Content Moderation following a User Report

Microsoft provides the information below in response to DSA Articles 15(1)(b)-(c) and 41(2)(a)-(b).

There are four ways to report harms on Microsoft Learn, based on the specific feature area where the harm occurred.

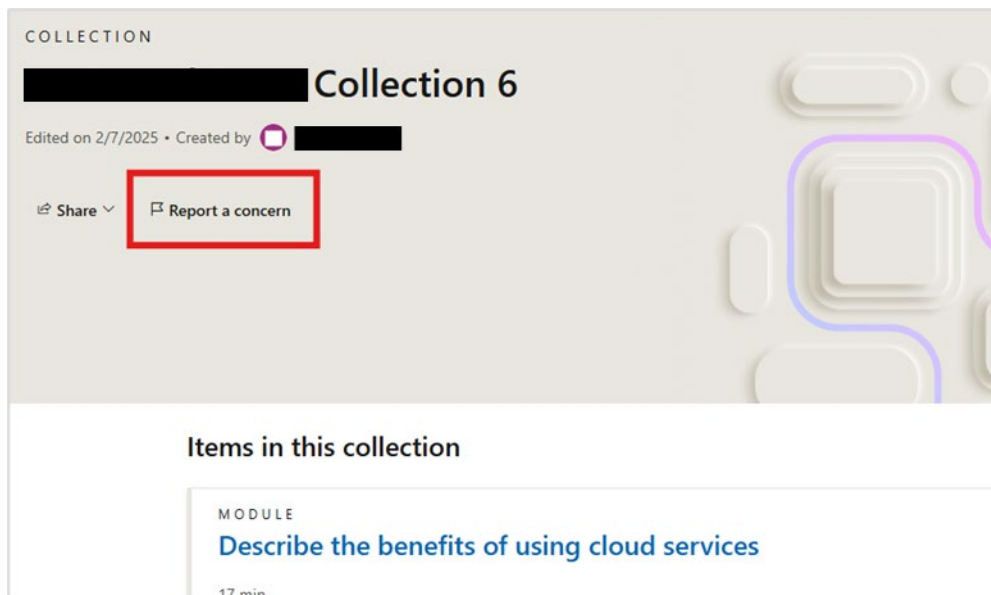
For a Q&A post:



1. All Q&A visitors see the **Report a Concern** button on any post/profile

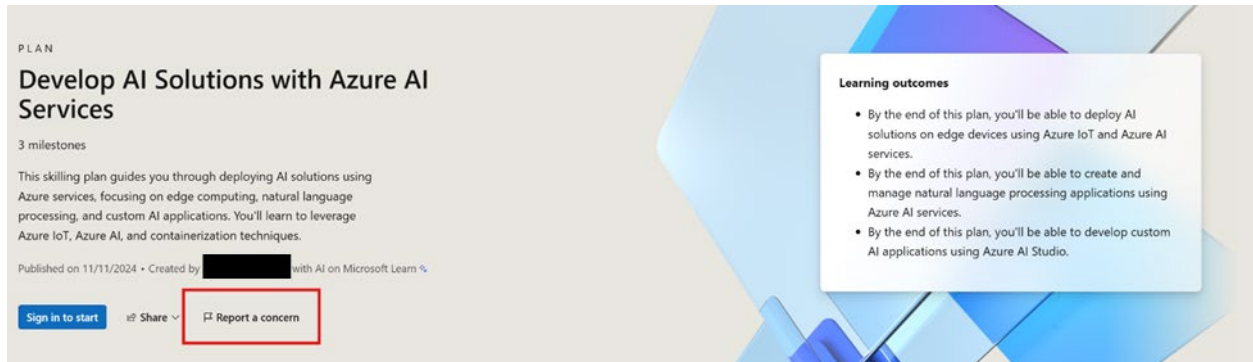
2. Users may select to report harms anonymously
3. After the harm report is submitted, users who reported the harm will receive email notification
4. After the harm report is reviewed by the moderator the user who submitted report will receive email notification about the action (unless they opted to stay anonymous)

For a Collection:



1. All Collection visitors see the **Report a Concern** button on any collection
2. Users may select to report harm anonymously
3. After the harm report is submitted, users who reported the harm will receive email notification
4. After the harm report is reviewed by the moderator the user who submitted report will receive email notification about the action (unless they opted to stay anonymous)

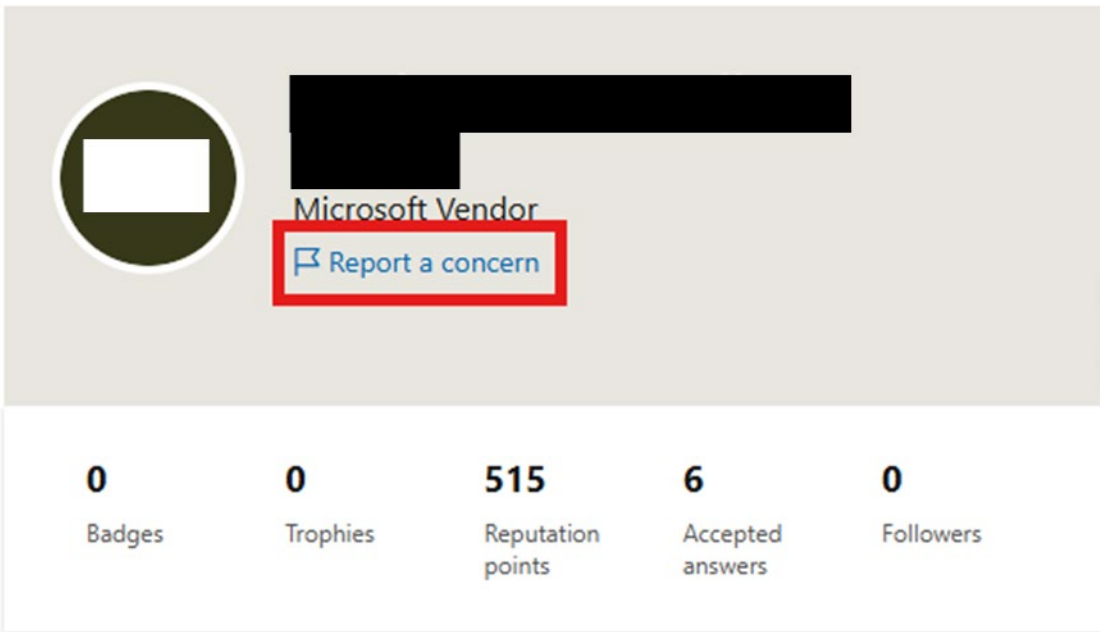
For a Plan:



1. All Plan visitors see the **Report a Concern** button on any plan
2. Users may select to report harm anonymously
3. After the harm report is submitted, users who reported the harm will receive email notification
4. After the harm report is reviewed by the moderator the user who submitted report will receive email notification about the action (unless they opted to stay anonymous)

For the AI-generated plan formation-based advisor input that ultimately produces a plan through AI, we utilize Azure OpenAI filters (LLM filters) to screen for harmful content. This process involves two levels of filtering at the model level to ensure content safety. The information is then stored in Azure Cosmos DB, and we provide dashboards to highlight this data. Additionally, we maintain a journey database to track all transactions related to the service. If a content filter is triggered, we log this event, and all related information is stored in Azure Cosmos DB.

For a Profile:



1. Each public profile has a **Report a concern** button below the Display Name
2. Users may select to report harm anonymously
3. After the harm report is submitted, users who reported the harm will receive email notification
4. The username and description is reviewed by a moderator who can take an action to take down the content using Profile DSA APIs.

Product Data: Microsoft Learn

Policy Violation	Number of notices received	Number of specific items of information included in the total number of notices	Median time to take action
Nonconsensual Behaviour	7	7	
Pornography or Sexualized Content	2	2	1.0
Protection of Minors	3	3	
Risk for public security	2	2	
Scope of Platform Services	6,883	6,883	14.6
Violence	4	4	

Content Moderation Appeals

Microsoft provides the information below in response to DSA Article 15(1)(d)

Microsoft did not receive notice of any content moderation appeals, which were applicable to Microsoft Learn, during the reporting period.

Government Requests

Microsoft provides the information below in response to DSA Article 15(1)(a).

Microsoft Learn received 0 (zero) orders from government authorities to act against alleged illegal content during the reporting period.

Member State	Total Government Requests	Request Granted
N/A	0	0